

Data Sheet – SEM9110(ICPZBS243QR)

A System-on-a-chip solution for 2.4 GHz IEEE 802.15.4

Applications

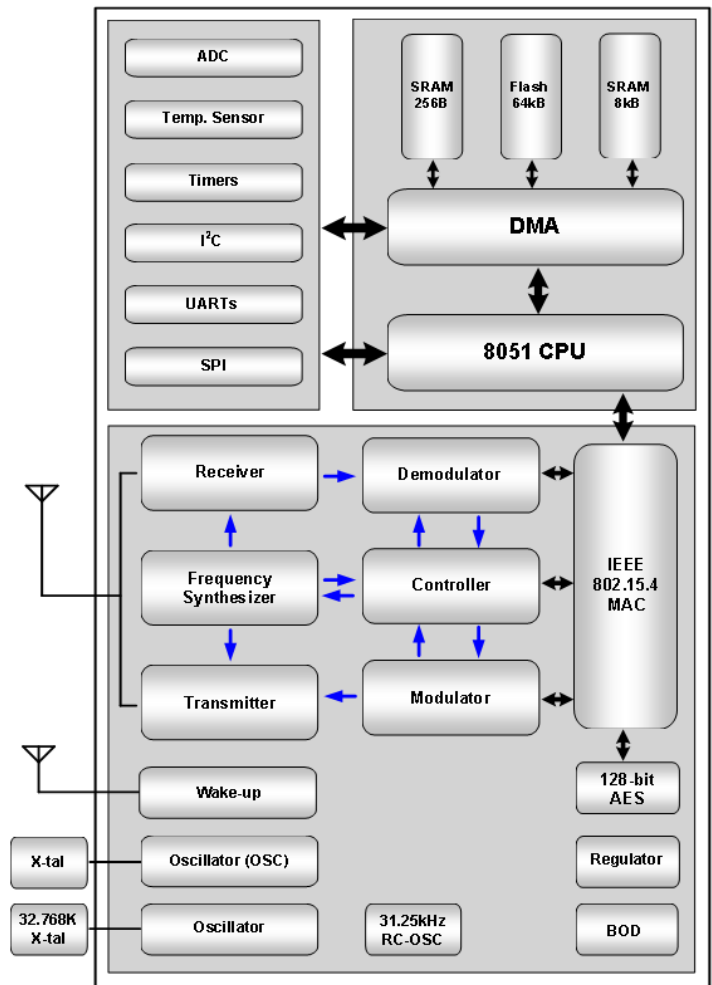
Low-Power wireless sensor network
 2.4 GHz IEEE 802.15.4 systems
 Home/building automation
 Industrial control and monitoring
 Home networks
 Consumer electronics
 Toy and gaming peripherals
 Telemetry and utilities

Overview

The SEM9110(ICPZBS243QR) is a low power-consuming, cost-efficient, system-on-a-chip device that supports the high-performance, low-power 8051CPU core, Flash, SRAM, peripherals, and the 2.4 GHz IEEE 802.15.4 standards.

The SEM9110(ICPZBS243QR) supports various operation modes to lower power consumption and is ideal for systems that require extremely low power consumption. Moreover

this has various peripherals such as UART, SPI, I²C, Timer, ADC, employs DMA, hardware MAC and AES encryption processor to minimize processor overhead for data processing.



Key Features

- Low power
 - RX: 26.5 mA, TX: 31 mA : (16MHz, CPU active)
 - Sleep current : 1.8 μ A (PM2, 31.25 kHz RC-OSC On, External Interrupt Available)
 - Three-step power control modes, two-step CPU power modes support
- Receiver sensitivity level : -100 dBm

- Modem data rate : 250kbps
- Transmit power +4 dBm
- Peripherals
 - ADC, Temp. Sensor, UART, SPI, I²C, Three timers, MAC timer, Watchdog timer, Sleep timer
 - Random Number Generator, 128-bit AES security processor
- H/W MAC support
- 64 kB in-system programmable flash memory & 1 kB information memory
- Digital RSSI/LQI support
- Battery monitor (LVI)
- Powerful DMA
- 17 general I/O pins
- JTAG S/W debugging
- 5×5 mm QFN 40-pin package
- High performance and low power 8051 CPU core
- 8 kB SRAM
- Short range wake-up functionality
 - Sensitivity : -17dBm

Datasheet Revision History

Date	Authors	Description
2012/06/22	SEMCO	First data sheet for released product.
2012/10/29	SEMCO	Update characteristics, specifications and fixed minor typing error.
2013/06/17	SEMCO	Update specifications and released v2.0.
2026/04/21	SKAIChips	Updated for the support of Zigbee GPD and OEPL.

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1. FEATURES

1.1. High-speed and Low-Power 8051 Compatible CPU

- Reduced instruction cycle time (up to 12 times fewer cycles compared with standard 8051), no External Memory Wait States (internally and externally triggered, separate for Program and Data Memory).
- Power Management Unit
- Program Memory (up to 64 kB)
- Program Memory Write Mode
- Control Unit
 - 8-bit instruction decoder
 - Reduced instruction cycle time (up to 12 times fewer cycles compared with standard 8051)
- Arithmetic-logic unit
 - 8-bit arithmetic operations
 - 8-bit logical operations
 - Boolean manipulations
 - 8x8 bits multiplication
 - 8/8 bits division
- Peripherals (17 I/O port : Function or GPIO)
 - ADC with two channel
 - 17 general I/O pins
 - UART
 - SPI
 - I²C
- On-chip JTAG Debug
- Six Timers
 - Mac timer
 - Three 16-bit timers
 - 32-bit watchdog timer, 20-bit sleep timer.
- Interrupt Controller
 - Two priority levels
 - 13 interrupt sources

1.2. 64 kB Flash Memory and 8 kB SRAM

Flash Memory

- Endurance: 20,000 cycles (min) with 20 ms erase and 20 μ s program time at 1.8 Vcc
- Memory Organization 64k x 8 + 1k x 8

- Page Erase Capability: 1 kbyte per page
- Greater than 100 years Data Retention under Room Temperature

SRAM

- 8 Kbytes of SRAM
- Completely Static Operation

1.3. Power Mode

The purpose of the Power management is to minimize the power consumption at a wireless battery powered communications device. The SEM9110(ICPZBS243QR) supports two power modes and two 8051 core power management modes. The SEM9110(ICPZBS243QR) is put into a power mode when no applications are active and when no tasks or interrupt routines need to be serviced. The SEM9110(ICPZBS243QR) provides power modes like power mode 2, 3. While in "power mode 2, 3" power of the system is shut down.

1.4. 802.15.4 Hardware MAC

- CRC-16 computation and checking over the MAC payload
- Hardware support for address recognition is included
- Hardware support for transmitting / receiving acknowledge frames is included
- Transmit data with CSMA-CA strobe is provided

1.5. Hardware AES support

- AES supported in hardware coprocessor

1.6. 2.4 GHz IEEE 802.15.4 Transceiver

In order to reduce the current consumption, the short range wake-up block is implemented. The wake-up circuit converts small RF input signals into a DC signal to trigger the interrupt. The circuit is composed of a passive voltage multiplying circuit, highly sensitive decision circuit, and over-voltage protection circuit. There are no power consumptions in this block because it is operated as passive components. When the SEM9110(ICPZBS243QR) operate in the power mode, utilizing this wake-up block is useful to wake up the system without any power consumption.

- Low-IF RF receiver
- Digital AGC, RSSI, LQI
- Direct Sequence Spread Spectrum (DSSS) transceiver
- 250kbps data rate support.
- Low current consumption
- High sensitivity (-100 dBm)
- Short range wake-up

1.7. Peripheral

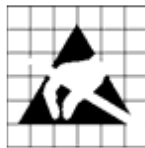
- Timer0, 1 : 8051 core inner timer. Consists of 16-bit registers.
- Timer2 (MAC Timer) : A 20-bit timer that acts as timer and BOS counter. Has a built-in scaler that has 2/4/8/16/32/64/128 as its value. Features Capture/Compare functions.
- Timer3 : A 16-bit timer that acts as timer. Has a built-in scaler that has 2/4/8/16/32/64/128 as its values. Features Capture/Compare functions.
- Watchdog Timer : 32-bit Timer.
- UART : Full-Duplex UART. High baud-rate accuracy.
- SPI : One synchronous Full-Duplex SPI. Master/slave mode.
- I²C : Master/slave mode. 7-bit addressing (extend 10-bit addressing). 100kbit/s ~ 400kbit/s rate.
- Powerful DMA Controller.
- Two channel ADC.
- Temperature Sensor
- LVI
- 17 configurable general-purpose digital I/O.

2. ABSOLUTE MAXIMUM RATINGS

In an environment where the Absolute Maximum Ratings as shown in Table 2.1 are exceeded, the device may suffer permanent damage.

Table 2.1 Absolute Maximum Ratings

Parameter		Min	Max	Units	Condition
Supply voltage		-0.3	3.6	V	
Input RF level			10	dBm	
Storage temperature range		-50	150	°C	
ESD	Human Body Model (HBM)		2	kV	RF pins & some VDD pins have a weak ESD protection. (RF_P, RF_N, VDD_RF, DVDD_18)
	Machine Model (MM)		200	V	
	Charged Device Model (CDM)		500	V	



Caution! ESD sensitive device. Precaution should be used when handling the device in order to prevent permanent damage.

3. OPERATING CONDITIONS

Operating conditions are shown in Table 3.1.

Table 3.1 Operating Conditions

Parameter	Min	Max	Units	Condition
Supply voltage for on-chip voltage regulator	2.2	3.6	V	
Operating ambient temperature range	-30	70	°C	

4. ELECTRICAL SPECIFICATIONS

Table 4.1 Electrical Specifications

Parameter	Min	Typ	Max	Units	Condition
RF Frequency Range	2400		2483.5	MHz	
Channel spacing		1		MHz	
Data rate		250k		bps	
Battery Monitor (LVI)					
Current consumption	40	50	80	μA	
Startup time			100	μs	
Setting time			2	μs	
Detection range	2.0		3.2	V	
Step size		40		mV	
Current consumption					
Power mode3		0.7		μA	16MHz
Power mode2		1.8		μA	16MHz
Receiver mode(CPU active)		26.5		mA	16MHz
Transmit mode(CPU active)		31		mA	16MHz
Carrier sense level	-78	-77	-76	dBm	
RSSI dynamic range		100		dB	
RSSI accuracy/linearity		±3		dB	
RSSI average time		128		μs	

Test condition: Vdd = 3V, Ta = 25°C

5. GENERAL CHARACTERISTICS

Table 5.1 General Characteristics

Parameter	Min	Typ	Max	Units	Condition
RX/TX turnaround		30	192	μs	

Test condition: Vdd = 3V, Ta = 25°C

6. RF

6.1. Receive Section

Table 6.1 Receive Section

Parameter	Min	Typ	Max	Units	Condition
Receiver Sensitivity 250kbps	-92	-100		dBm	PER=1 % with 22 bytes
Saturation (maximum input level)	0	10		dBm	
Adjacent channel rejection +5Mhz channel spacing -5Mhz channel spacing		39 35		dB dB	Wanted signal -82 dBm, IEEE 802.15.4 modulated signal at +/-5 MHz, PER = 1 %
Alternate channel spacing +10Mhz channel spacing -10Mhz channel spacing		49 49		dB dB	Wanted signal -82 dBm, IEEE 802.15.4 modulated signal at ± 10 MHz, PER = 1 %
Channel rejection $\geq +15$ Mhz		52		dB	Wanted signal @ -82 dBm. IEEE 802.15.4 modulated signal at $\geq +15$ MHz channel spacing, PER = 1%
Spurious emission 30-1000 Mhz 1-12.75 Ghz			-70 -53	dBm dBm	
Frequency error tolerance 250 kbps	-300		300	kHz	
Symbol rate error tolerance 250 kbps	-120		120	ppm	
Dynamic range	-100		10	dBm	
Dynamic range step		1		dB	

Test condition: Vdd = 3V, Ta = 25°C

6.2. Transmit Section

Table 6.2 Transmit Section

Parameter	Min	Typ	Max	Units	Condition
Output power					
0dBm Setting	-3	0		dBm	
Max power Setting		4		dBm	
Programmable output power range		40		dB	
Output power control step		4.5~6		dB	
Harmonics (Max power unmodulated signal) ⁽¹⁾					
2 nd harmonic		-40		dBm	
3 rd harmonic		-59			
Spurious emission (Max power modulated signal) ⁽²⁾					
30-1000 MHz		-65		dBm	
1-12.75 GHz (outside restricted bands)		-40		dBm	
1.8-1.9 GHz		-65		dBm	
5.15-5.3 GHz		-70		dBm	
EVM(Error Vector Magnitude)		10	25	%	CPU active @16MHz
@ Max power modulated signal		15	25	%	CPU active @32MHz
Optimum load impedance		50		Ω	

Test condition: Vdd = 3V, Ta = 25°C

(1), (2) is measured using a simple low-pass filter connected between matching network and RF connector (1 nH in parallel 0.7pF).

7. OSCILLATOR

7.1. 16 MHz & 32 MHz Crystal Oscillator

By setting internal conditions you can selectively use either 16 MHz or 32 MHz X-tal oscillator.

Table 7.1 16 MHz Crystal Oscillators

Parameter	Min	Typ	Max	Units	Condition
Crystal frequency		16		MHz	
Crystal frequency accuracy requirement	-40		40	ppm	
Start-up time			2	ms	
Capacitance load	10	16	22	pF	

Test condition: Vdd = 3V, Ta = 25°C

Table 7.2 32 MHz Crystal Oscillators

Parameter	Min	Typ	Max	Units	Condition
Crystal frequency		32		MHz	
Crystal frequency accuracy requirement	-40		40	ppm	
Start-up time			2	ms	
Capacitance load	10	16	22	pF	

Test condition: Vdd = 3V, Ta = 25°C

7.2. Low Frequency RC Oscillator

Table 7.3 Low Frequency RC Oscillator

Parameter	Min	Typ	Max	Units	Condition
Calibrated frequency		31.25		kHz	
Frequency accuracy after calibration			±1	%	
Initial calibration time			10	ms	

Test condition: Vdd = 3V, Ta = 25°C

The SEM9110(ICPZBS243QR) has a low-power, low-frequency RC oscillator that runs all the time. Its calibrated frequency is 31.25 kHz. The RC oscillator is calibrated automatically by setting the modem register RC_32K_CAL 1.CAL_ON (0xDFF0).

RC Oscillator Calibration Block Diagram as below:

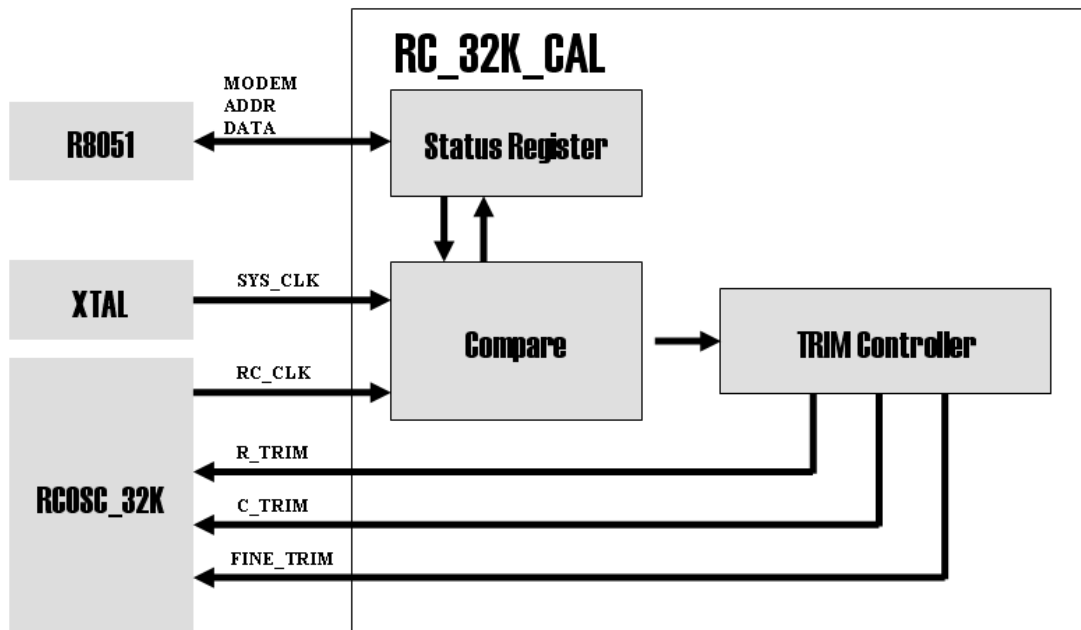


Figure 7.1 RC Oscillator Calibration Block Diagram

7.3. High Frequency RC Oscillator

Table 7.4 High Frequency RC Oscillator

Parameter	Min	Typ	Max	Units	Condition
Calibrated frequency	12	32	40	MHz	
Frequency accuracy after calibration			±0.5	%	
Initial calibration time			500	us	

Test condition: Vdd = 3V, Ta = 25°C

The SEM9110(ICPZBS243QR) has a internal high-frequency RC oscillator that runs when sleep to wakeup, it support to oscillate external crystal oscillator more fast.. Its calibrated frequency is determined by system clock. The high-frequency RC oscillator is calibrated by setting HFRC_CAL_ON to '1' and setting PD_CLKF to '0' at modem register RC_32M_CAL (0xDFF6).

8. FREQUENCY SYNTHESIZER CHARACTERISTICS

Table 8.1 Frequency Synthesizer Characteristics

Parameter	Min	Typ	Max	Units	Condition
IF frequency		2		MHz	
Crystal oscillator frequency		16 or 32		MHz	
Crystal frequency accuracy req.	-40		40	ppm	
Crystal OSC start-up time		0.86		ms	
PLL phase noise	-107	-105		dBc	@ 1 MHz (/Hz)
PLL Frequency range	2200	2400	2500	MHz	5 MHz ch sp
PLL loop bandwidth		100		kHz	
PLL lock time		50		μs	

Test condition: Vdd = 3V, Ta = 25°C

9. ADDITIONAL FUNCTION BLOCKS

9.1. Short Range Wake-up

Table 9.1 Wake-up Characteristics

Parameter	Min	Typ	Max	Units	Condition
Short range wake-up Sensitivity		-17		dBm	

Test condition: Vdd = 3V, Ta = 25°C

9.2. Brownout Detector

Table 9.2 Brownout Detector Characteristics

Parameter	Min	Typ	Max	Units	Condition
BOD falling Reset Voltage	1.7	1.72	1.77	V	
BOD rising Reset Voltage	1.84	1.86	1.91	V	

Test condition: Vdd = 3V, Ta = 25°C

9.3. Analog Temperature Sensor

Table 9.3 Temperature Sensor Characteristics

Parameter	Min	Typ	Max	Units	Condition
ADC output code at -40°C		69		Dec.	Common SFR setting : TSCN(0xE7) : 0x22, TSTRIM(0xE6) : 0x55
ADC output code at 0°C		236		Dec.	
ADC output code at 30°C		510		Dec.	
ADC output code at 70°C		1020		Dec.	
Temperature coefficient		10		Dec./°C	ADC output
Error in calculated temperature	-6		6	°C	
Current consumption			650	μA	

Test condition: Vdd = 3V, Ta = 25°C

10. DC CHARACTERISTICS

DC characteristics of the SEM9110(ICPZBS243QR) are shown in Table 10.1.

Table 10.1 DC Characteristics

Parameter	Min	Typ	Max	Units	Condition
Logic “0” input voltage	0		0.7	V	
Logic “1” input voltage	VDD - 0.7		VDD	V	
Logic “0” output voltage	0		0.5	V	
Logic “1” output voltage	VDD - 0.3		VDD	V	
Logic “0” input current	N / A	-1	-1	μA	
Logic “1” input current	N / A	1	1	μA	

Test condition: Ta = 25°C

11. PIN CONFIGURATION

11.1. QFN 40-pin Package

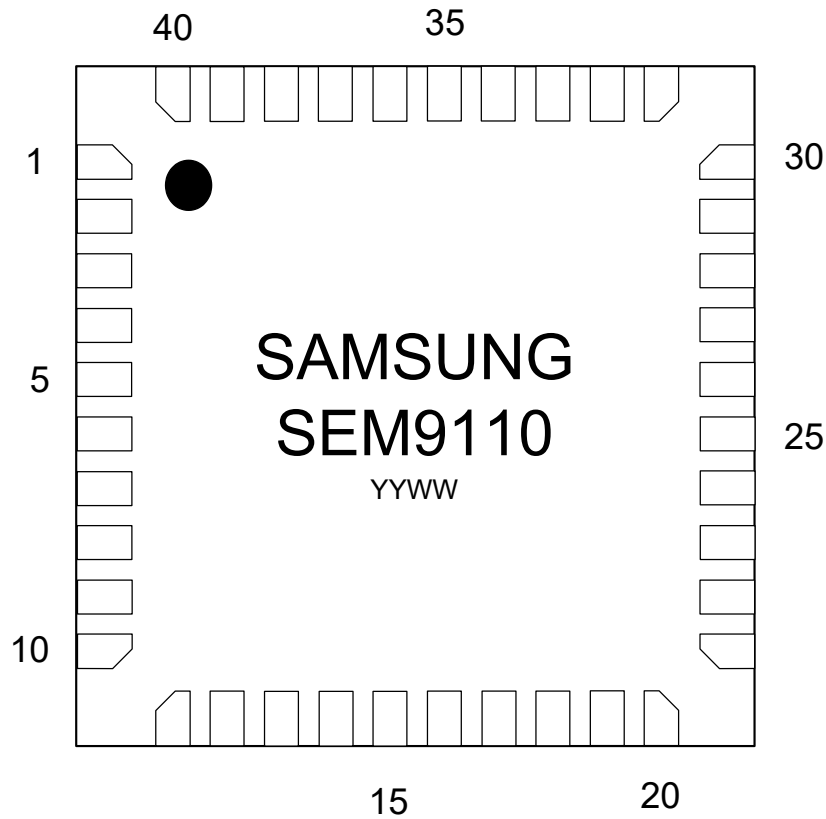


Figure 11.1 QFN 40-Pin Assignment

Table 11.1 QFN 40-pin Package Overview

Package Pin No.	Name	Type	Description
1	VDD_RF	P	RF Voltage Supply Input(2.0V)
2	GND_RF	G	RF Ground
3	RF_N	RF	RF In / Out Port (differential)
4	RF_P	RF	RF In / Out Port (differential)
5	VDD_BB / VR_D_OUT	P	Transceiver IF (Filter / ADC / DAC) / Voltage Supply 1.8V / Digital Regulator Voltage Out 1.8V
6	GNDD33	G	Digital Voltage Regulator Ground
7	VDDD33	P	Digital/Sleep Regulator Voltage Supply (3V)
8	REXT	A	Reference Current Generation Resistor 24Kohm ($\pm 1\%$)

9	VR_S_OUT	P	Sleep Regulator Voltage Out(1.9V)
10	RF_WU	A	Short Range RF Wake-up input
11	XTALI	A	Crystal 16 / 32 MHz Input Port
12	XTALO	A	Crystal 16 / 32 MHz Output Port
13	VDD_MCU	P	Digital CPU Power Supply 1.8V
14	P0.0	DIO	GPIO P0.0(SPI SCK)
15	P0.1	DIO	GPIO P0.1(SPI MOSI)
16	P0.2	DIO	GPIO P0.2(SPI MISO)
17	P0.3	DIO	GPIO P0.3(SPI SS)
18	LF_XTAL_IN	AIO	LF X-tal Input(32.768kHz)
19	LF_XTAL_OUT	AIO	LF X-tal Output(32.768kHz)
20	P0.6	DIO	GPIO P0.6(UART TX)
21	P0.7	DIO	GPIO P0.7(UART RX)
22	RESET_N	DI	Global Reset Input (Active Low)
23	P1.0	DIO	GPIO P1.0
24	P1.1	DIO	GPIO P1.1
25	P1.2	DIO	GPIO P1.2
26	VDD3D2	P	Digital Power Supply 3V for PAD
27	P1.3	DIO	GPIO P1.3
28	P1.4	DIO	GPIO P1.4(I ² C SCL)
29	P1.5	DIO	GPIO P1.5(I ² C SDA)
30	P1.6	DIO	GPIO P1.6
31	P1.7	DIO	GPIO P1.7(Debugger TCK)
32	P2.0	DIO	GPIO P2.0(Debugger TDI)
33	P2.1	DIO	GPIO P2.1(Debugger TDO)
34	P2.2	DIO	GPIO P2.2(Debugger TMS)
35	ADC1	AIO	ADC ch1 Input
36	ADC2	AIO	ADC ch2 Input
37	REF_OUT	AIO	Reference Voltage Output(1.8V)
38	VDD_DIG	P	Digital Modem Power Supply 1.8V
39	VDD_GUARD / VDDA33	P	Guard-ring VDD Node / Analog Regulator Voltage Supply 3V
40	VR_OUT	P	Analog Regulator Output (2.0V)

12. CIRCUIT DESCRIPTION

Request Reference Manual for this section

13. APPLICATION CIRCUIT

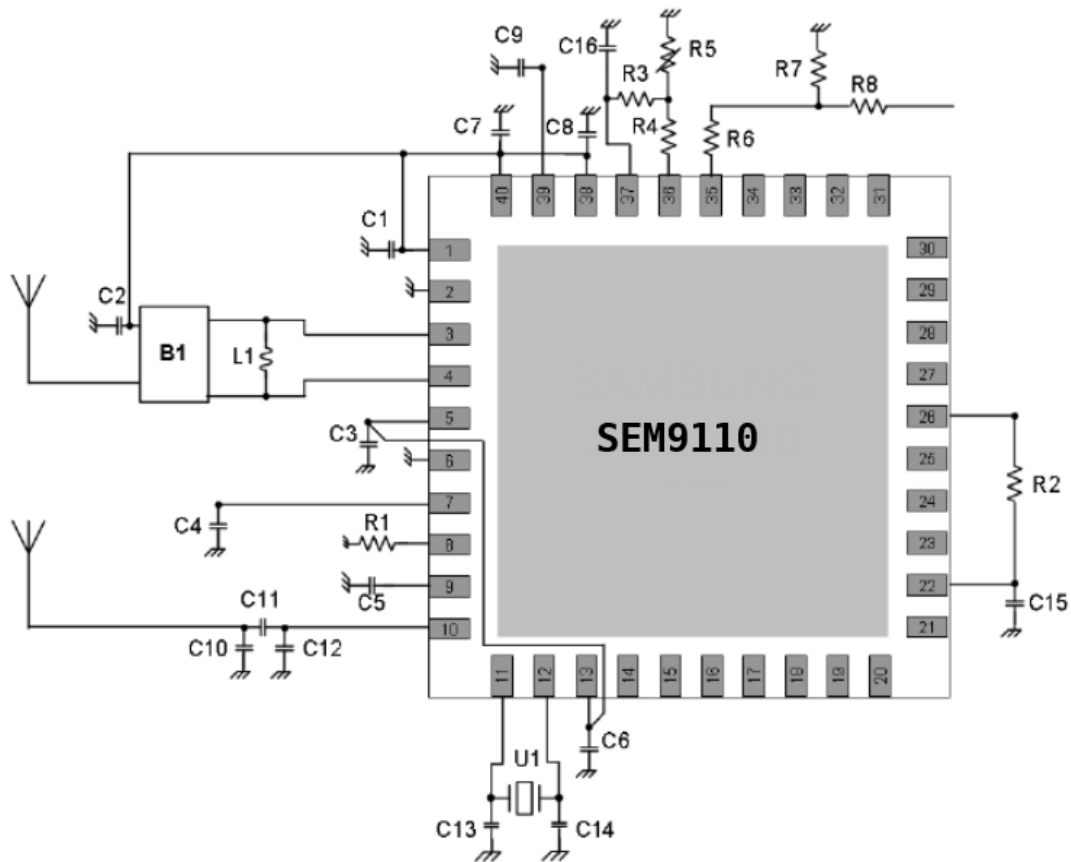


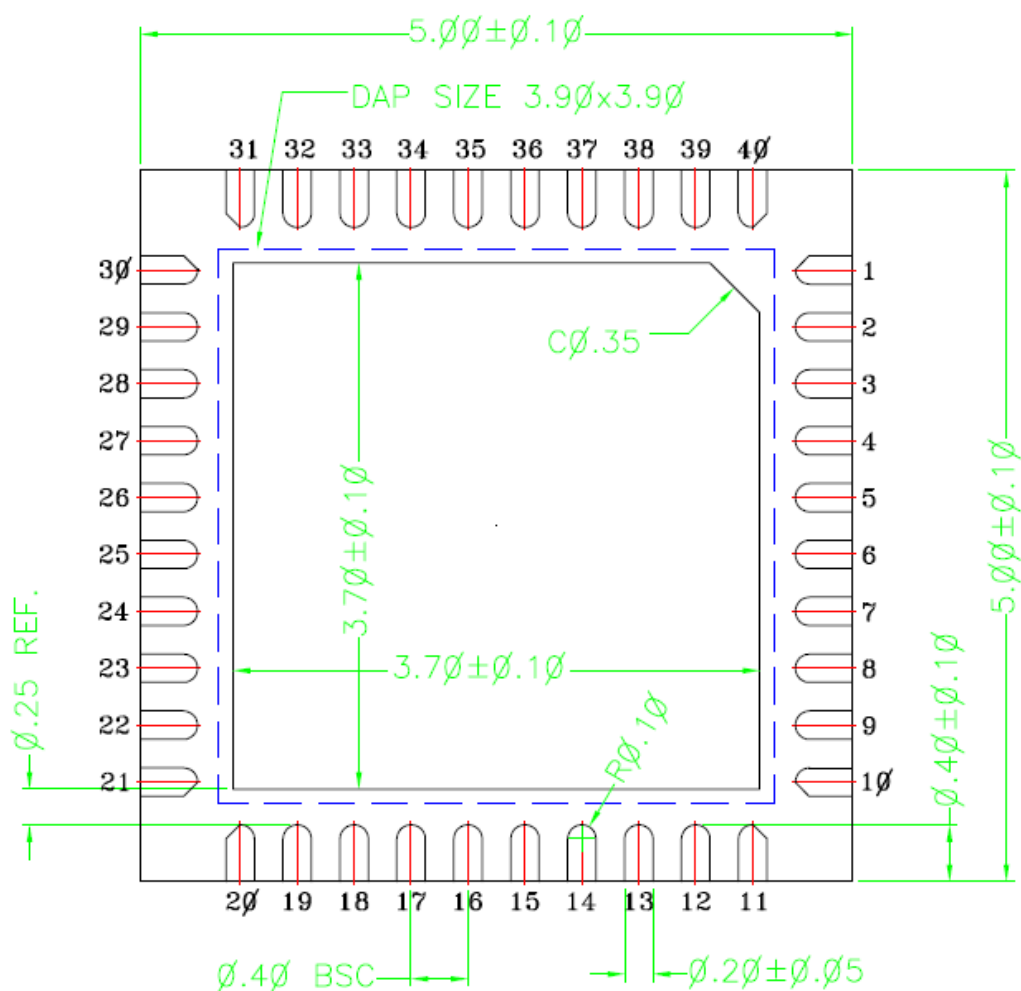
Figure 13.1 QFN 40-pin Package Typical Application Circuit

Table 13.1 QFN 40-pin Package Overview of External Components

Ref	Description
B1	Balun
C1,C2,C3,C4,C5,C6,C7,C8,C9	Bypass capacitor
C10,C11,C12	RF Matching capacitor
L1	Matching & DC feed inductor
C13,C14	Crystal load capacitor
U1	16 MHz or 32 MHz x-tal oscillator
R1	Resistor for current reference
R2,C15	R, C for reset
R3,R4,R5,C16	R, C, Thermistor for Temp. Sensor
R6,R7,R8	R for Detecting the Charge Pump Output

14. PACKAGE DESCRIPTION

POD IN BOTTOM VIEW



POD IN SIDE VIEW

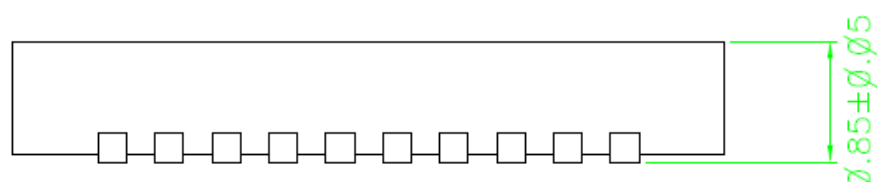


Figure 14.1 QFN 40-pin Package Dimensions Drawing (unit mm)

15. PCB AND STENCIL DESIGN RECOMMENDATIONS FOR TAPP AND LPCC PACKAGE

15.1. QFN 40-pin PCB Land Pattern Dimensions

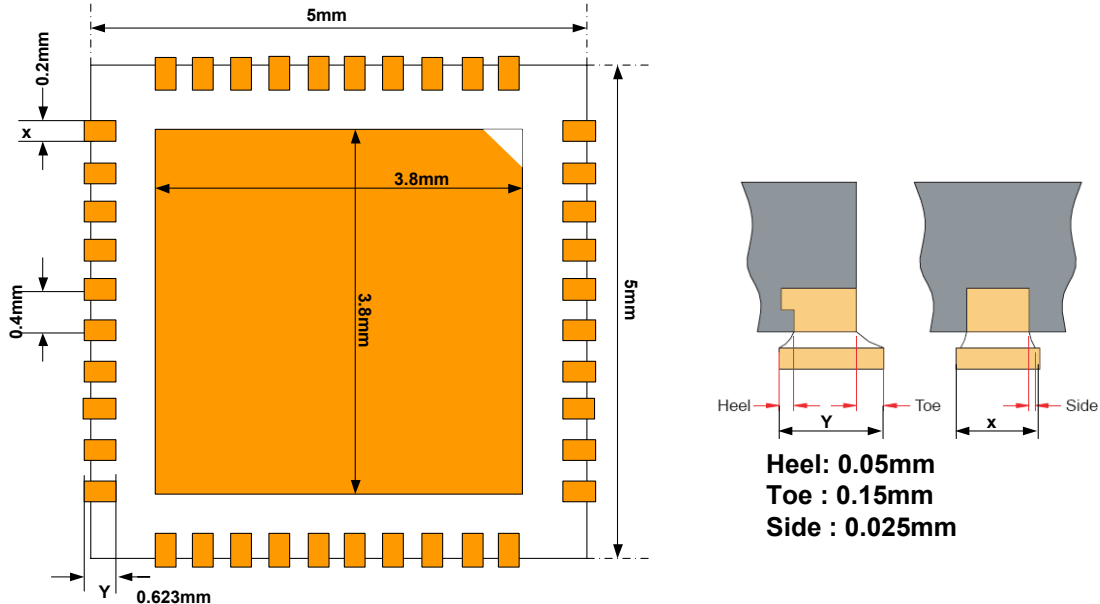


Figure 15.1 Recommended PCB Land pattern for QFN 40-pin Package

15.2. QFN 40-pin Stencil Design Recommendation

Stainless steel stencils are recommended for solder paste application. A stencil thickness of 0.125 mm ~ 0.15 mm is recommended for screening. For improve release, the aperture walls should be trapezoidal and the corners rounded. Solder paste should be printed on the PCB thermal land by designing an array of smaller openings that sum up to ~50% of the thermal pad area (see the figure 15.2). The shape of the smaller stencil opening is optional.

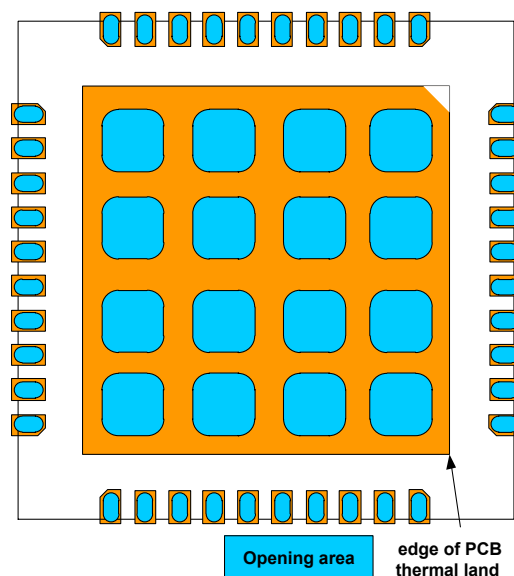


Figure 15.2 Recommended Stencil design for QFN 40-pin Package

16. REFLOW PROFILE

The packages are attached to the PCB by infrared or convection mass reflow techniques as part of standards SMT processing.

Thermal profiling of convection / IR reflow machine is required for each product design.

- For Pb-free solder, the reflow temperature shall not exceed 260 °C with time above liquids temperature(217 °C) of 45 seconds minimum.
- Standard CDA atmosphere is acceptable but the use of nitrogen is recommended.
- The airflow may need to be reduced in some cases to prevent the lightweight parts from shifting or being blown off. The temperature would have to be adjusted accordingly to maintain the reflow profile.
- Do not exceed the maximum motherboard temperature recommended by the supplier.

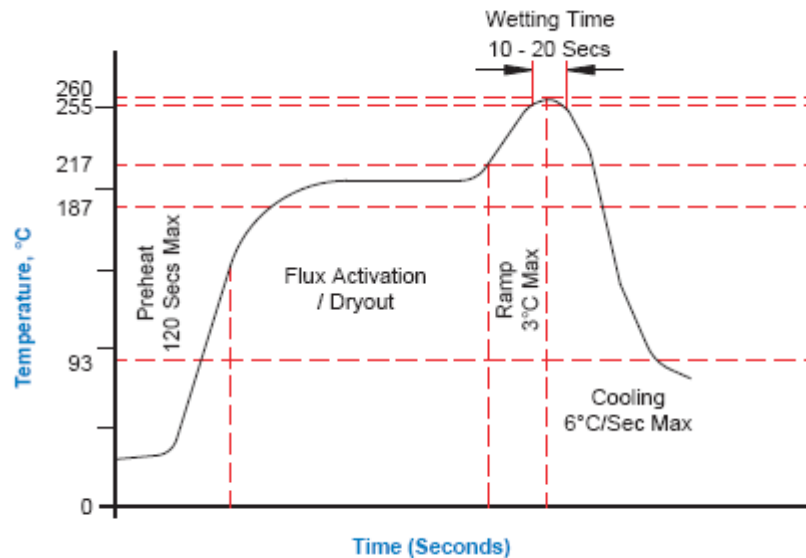


Figure 16.1 Pb-free Solder Reflow Profile

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